Quantum Convolutional Codes Constructed from Classical Self-Orthogonal Convolutional Codes

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***Abstract*—Quantum convolutional codes (QCCs) are a promising class of codes for protecting continuous quantum information streams. In this paper, we construct new QCCs from classical self-orthogonal convolutional (CSOC) codes. We develop a methodology that achieves the required code orthogonality and allows for code performance optimization, generalizing to the full algebraic structure. This approach leverages the significant body of work in classical CSOCs to achieve a specified free distance.**

***Index Terms*—Quantum convolutional codes, stabilizer codes, free distance, symplectic orthogonality.**

I. Introduction

QUANTUM error correction (QEC) is a critical component for the reliable execution of quantum computation and long-distance quantum communication in noisy environments [1]. The stabilizer formalism introduced in [1] provides a unifying framework to construct and analyze quantum codes, generalizing classical linear codes to the Pauli-operator setting. Many types of quantum error-correcting codes, including Calderbank–Shor–Steane (CSS) codes and topological surface codes, fit naturally into this stabilizer framework [2].

Quantum *convolutional* codes (QCCs) [3], [4] extend stabilizer-based quantum codes to protect streams of qubits, drawing inspiration from classical convolutional coding principles. As in the classical case, QCCs incorporate memory elements and allow online encoding and decoding, making them attractive for continuous quantum communication scenarios [3], [5], [6]. One of the foremost challenges for QCCs is to ensure that a finite number of physical qubit errors does not lead to unbounded error propagation. This requirement is tied to the notion of *non-catastrophic* encoding, which has been characterized through polynomial and symplectic representations [3], [7].

To assess how well a QCC can detect and correct errors, one typically uses distance properties (notably the *free distance*) and associated weight enumerators. A key focus involves effectively terminating the infinite convolutional structure without undermining the stabilizer constraints. A *tail-biting* approach addresses this by wrapping the memory state of the final frame back to match the initial state, preserving symplectic orthogonality and yielding a valid finite block code, see, e.g., [3], [4].

II. A Self-Orthogonal Construction Theorem for Quantum Convolutional Codes

We present a theorem that provides a valid construction of high-performance quantum convolutional codes by leveraging the structure of classical convolutional self-orthogonal codes (CSOCs). The theorem establishes both the validity of the quantum code and the direct inheritance of the free distance of the classical code. We follow [8] to obtain the CSS construction.

Theorem 1. *Let C be a classical rate k/n convolutional code with a corresponding $(n-k)\times n$ polynomial parity-check matrix $H_C(D)$. Let Q a quantum convolutional code be defined by the $(n-k)\times 2n$ stabilizer matrix $H_Q(D) = \left[\, Z(D) \mid X(D) \right]$, where $Z(D) = H_C(D)$ and $X(D) = H_C(D)$. Then:*

1) ***Validity:*** *The code Q is a valid QCC, satisfying the symplectic orthogonality condition.*
2) ***Distance Preservation:*** *The free distance of the quantum code Q, denoted $d_f(Q)$, is equal to the free distance of the classical code C, denoted $d_{free}(C)$.*

Proof. To prove that Q is a valid quantum convolutional code, we must demonstrate that its stabilizer matrix satisfies the symplectic orthogonality condition, which ensures that all stabilizer generators commute. This is given by [**?**]: $X(D)\,Z(D^{-1})^T + Z(D)\,X(D^{-1})^T = 0 \pmod 2$. Substitute $X(D) = H_C(D)$ and $Z(D) = H_C(D)$ into the above, then trivially

$$H_C(D)\,H_C(D^{-1})^T + H_C(D)\,H_C(D^{-1})^T = 0.$$

The free distance of a QCC, $d_f(Q)$, is determined by the minimum weight of undetectable errors. An error is undetectable if it commutes with all stabilizer generators. The minimum weight undetectable error will be either a pure-Z error pattern or a pure-X error pattern. The free distance is therefore the smaller of the minimum weights of these two types of error patterns, $d_f(Q) = \min\left(d_X^{\perp},\, d_Z^{\perp}\right)$.

In this case:

- $d_Z^{\perp}$ is the minimum weight of a non-zero polynomial vector $u(D)$ corresponding to a pure-Z error that is undetected by the X-part of the stabilizer. This condition is $u(D)\,X(D^{-1})^T \;=\; 0$.

- $d_X^{\perp}$ is the minimum weight of a non-zero polynomial vector $v(D)$ corresponding to a pure-X error that is undetected by the Z-part of the stabilizer. This condition is $v(D)\,Z(D^{-1})^T \;=\; 0$.

In the construction under consideration, we have set $X(D) = Z(D) = H_C(D)$. This immediately implies that the conditions for an undetectable Z-error and an undetectable X-error become identical:

- The condition for an undetectable Z-error is

$$u(D)\,H_C(D^{-1})^T \;=\; 0.$$

- The condition for an undetectable X-error is

$$v(D)\,H_C(D^{-1})^T \;=\; 0.$$

The sets of vectors $\{u(D)\}$ and $\{v(D)\}$ are therefore the same, and consequently their minimum weights must be equal, i.e., $d_X^{\perp} \;=\; d_Z^{\perp}$. $\square$

Corollary 1 (Optimizing $Z(D)$ for maximum distance)**.** *Let $X(D)$ be a fixed $r \times n$ binary polynomial matrix (defining the X component of the r stabilizer generators). Assume $X(D)$ is non-catastrophic and covers all qubits (each column has at least one 1 in X, so that no single-qubit Z error is immediately undetectable by default). Let $d_X^{\perp}$ denote the minimum Hamming weight of any nonzero polynomial vector $u(D)$ orthogonal to all rows of $X(D)$ (i.e., the minimum weight of U_X defined above). That is, $d_X^{\perp} = \min_{u \in U_X} \mathrm{wt}(u)$. This is effectively the minimum distance of the classical dual code of X's row-space. Let $d_Z^{\perp}$ analogously denote the minimum weight of U_Z (the dual of the chosen Z). Then:*

1. *For any valid choice of $Z(D)$, the quantum free distance is bounded by*

$$d_{\mathit{free}} \le \min\{d_X^{\perp}, d_Z^{\perp}\}.$$

   In particular, $d_{\mathit{free}} \le d_X^{\perp}$, with equality only if $Z(D)$ is chosen such that no Z-error of weight $d_X^{\perp}$ commutes with all X (i.e. $d_Z^{\perp} \ge d_X^{\perp}$). Likewise, $d_{\mathit{free}} \le d_Z^{\perp}$, with equality only if $d_X^{\perp} \ge d_Z^{\perp}$.
2. *There exists a choice of $Z(D)$ (satisfying all symplectic constraints) that achieves the maximum possible free distance $d_{\mathit{free}} \;=\; \min\{d_X^{\perp}, d_Z^{\perp}\}$, by balancing the detectability of X and Z errors. In fact, one can always choose $Z(D)$ such that $d_Z^{\perp} \ge d_X^{\perp}$ (so that $d_{\mathit{free}} = d_X^{\perp}$), or vice versa if $d_Z^{\perp}$ were inherently larger.*
3. *Among all $Z(D)$ that achieve this maximal distance, one can further select $Z(D)$ to minimize the generator weights (i.e. the Hamming weight of each row polynomial). This can be done by performing row operations and column permutations on the stabilizer (which preserve commutation and distance) to arrive at a weight-reduced but equivalent $Z(D)$. Thus, an optimal $Z(D)$ can be chosen to simultaneously maximize d_{free} and to use the smallest possible support on each generator.*

NO	$G(D)$	g_{11}	g_{12}	g_{13}	g_{21}	g_{22}	g_{23}	d_{free}
1	$X(D)$	010	011	011	001	010	011	4
	$Z(D)$	110	101	100	101	100	100	
2	$X(D)$	0110	0100	0010	1000	0010	1110	4
	$Z(D)$	0100	1101	1000	0110	1110	0000	
3	$X(D)$	110	001	101	001	110	111	5
	$Z(D)$	111	111	000	111	111	000	
4	$X(D)$	000011	000111	000110	001011	000000	001111	7
	$Z(D)$	111000	101100	110000	101100	110100	100000	
6	$X(D)$	011	110	111	100	101	101	5
	$Z(D)$	100	111	010	101	011	010	
7	$X(D)$	0001100	0010110	0010110	0011010	0001000	0011010	7
	$Z(D)$	1000000	1011101	1000000	0011110	0010110	0000000	
7	$X(D)$	000011	000110	000111	001110	001111	000001	7
	$Z(D)$	011110	010110	010000	111010	101000	100000	
8	$X(D)$	0001010	0010100	0011010	0010100	0001110	0011010	7
	$Z(D)$	1011010	1001111	1000000	0011101	0110010	0000000	

TABLE I

QUANTUM CONVOLUTIONAL CODES WITH FIXED $X(D)$.

III. Numerical Results

Theorem 1 used the classical parity-check matrix $H_C(D)$ of a convolutional self-orthogonal code (CSOC) with the quantum stabilizer matrix $H_Q(D) = [\,X(D)\,|\,Z(D)\,]$ that is obtained by placing the same $H_C(D)$ in both Pauli blocks. Because $H_C(D)$ is self-orthogonal, the symplectic condition $X(D)Z(D^{-1})^T + Z(D)X(D^{-1})^T = 0$ is satisfied identically, so every quantum convolutional code (QCC) preserves both the rate and the free distance of its classical parent, i.e., $d_f(Q) = d_{\mathrm{free}}(C)$.

Table I lists eight representative QCC pairs obtained following a computer search according to Corollary 1. Their measured free distances are $d_f = \{4,\, 4,\, 5,\, 7,\, 5,\, 7,\, 7,\, 7\}$, exactly matching the classical values and confirming the theoretical prediction of Corollary 1. Standard tail-biting termination over a span proportional to the encoder memory converts each convolutional code into a finite-length block code while preserving the same distance, providing practical encoders for storage and communication systems.

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